



NCCER Module 22106



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Grades





Section Two

Performing Site Measurements

Objective

2. Describe the equipment and methods used in making site measurements and establishing finish grades.
 - a. Explain how to make horizontal and vertical measurements.
 - b. Describe the electronic equipment and systems used in site measurement and grading.
 - c. Explain how to establish and check finish grade.



Performance Tasks

Under the supervision of an instructor, trainees should be able to do the following:

2. Check horizontal and vertical distances of cut and fill slope stakes.
3. Check finish subgrade on a cross slope.



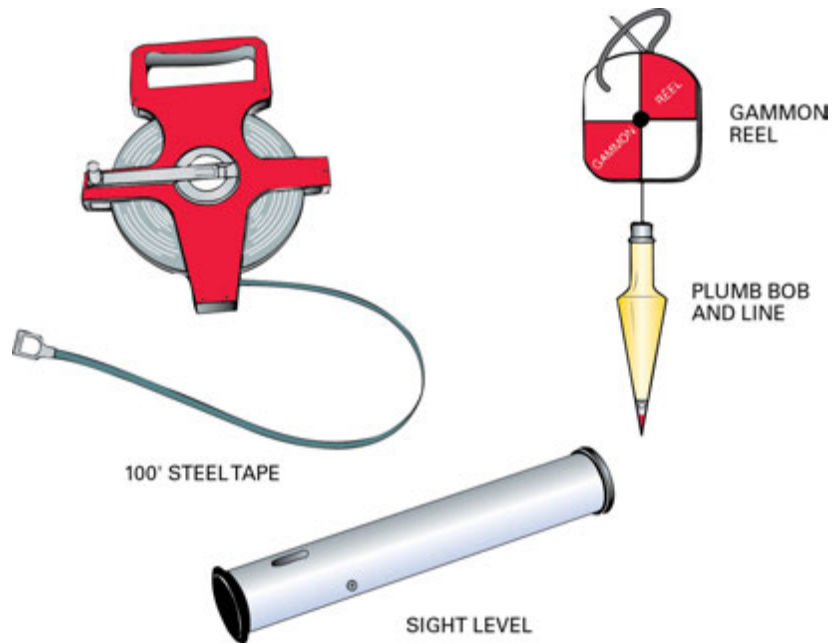
2.0.0 Performing Site Measurements

- Heavy equipment machines are often equipped with some type of **global positioning system (GPS)**.
- The GPS/survey equipment is aligned to the control points positioned by the survey team.



2.1.0 Performing Site Measurements (1 of 2)

- The sight level can be used to measure vertical distances.
- The measuring tape measures horizontal distances.
- The plumb bob establishes a vertical line.
- Surveyors use these devices only for rough grading.



2.1.0 Performing Site Measurements (2 of 2)

- Using a tripod gives the leveling device better stability.
- A leveling rod is used with a sight level or an automatic level to measure vertical distances.



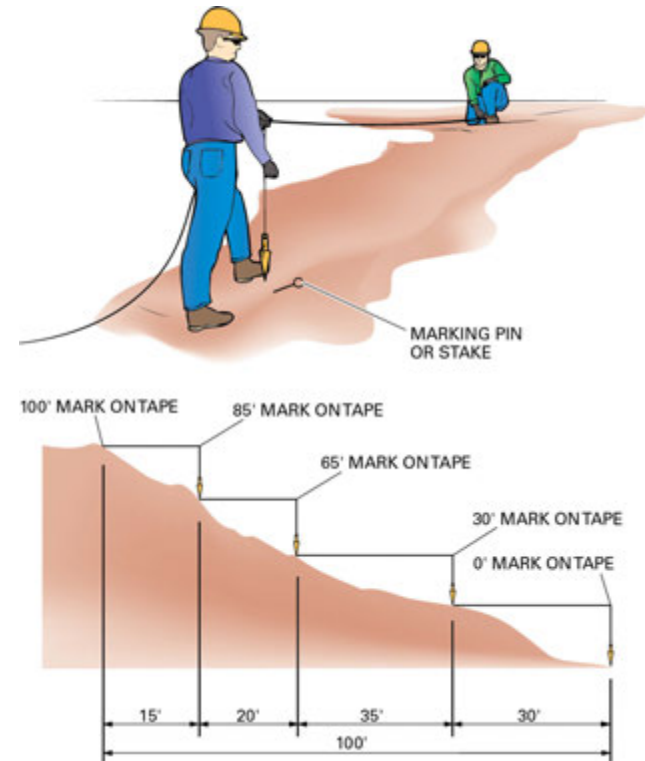
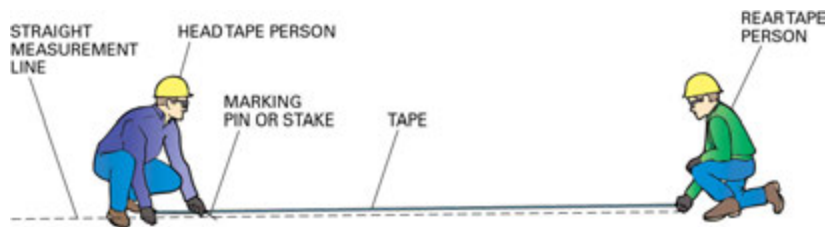
LENKER ROD
(SAN FRANCISCO STYLE)



TELESCOPING
ROD

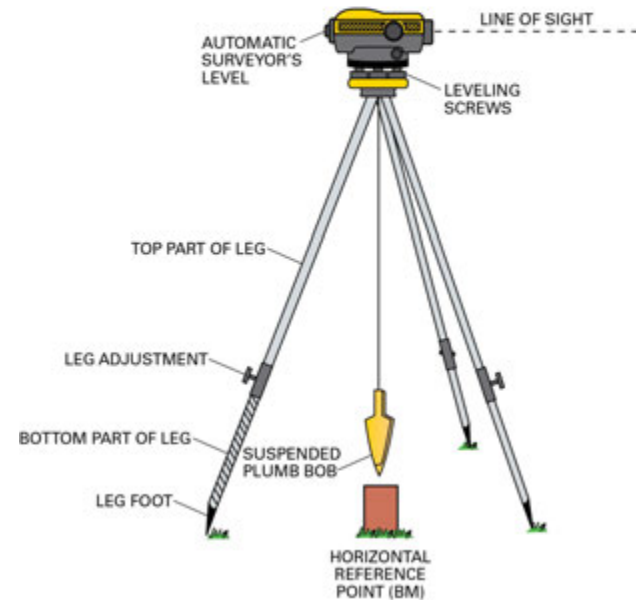
2.1.1 Horizontal Distance Measurements

When measuring a horizontal distance across a steep slope, measure the distance in a series of steps while holding the tape horizontal (level) as shown.



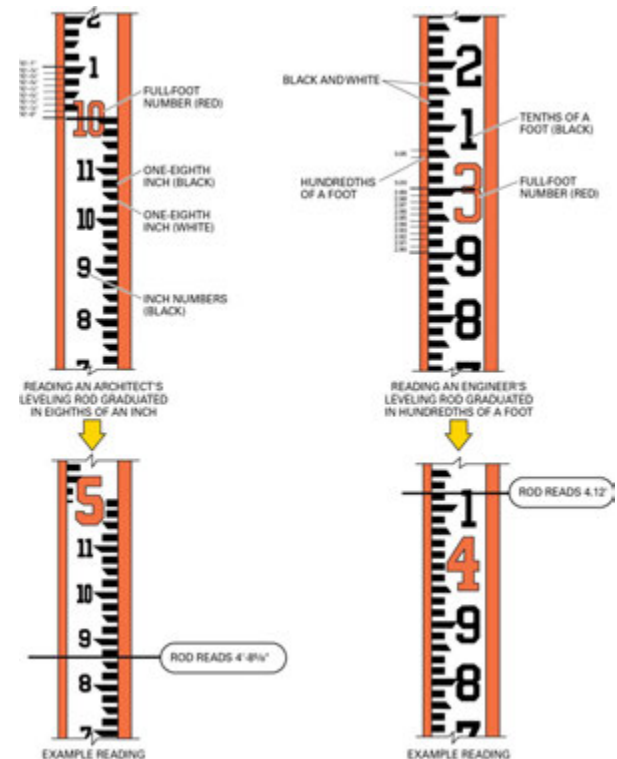
2.1.2 Vertical Distance Measurements (1 of 4)

- Differential leveling permits setting grade stakes and checking slopes after the grading work is done.
- Vertical distances may be measured using a sight or automatic level and leveling rod.
- Any movement of the tripod during operation requires repeating all measurements.
- Surveying methods require double-checking each reading.



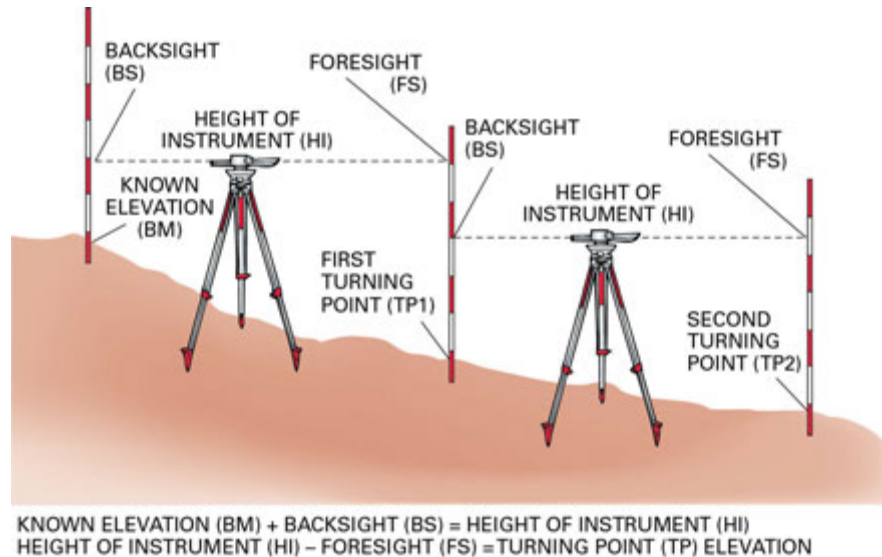
2.1.2 Vertical Distance Measurements (2 of 4)

- The architect's rod, measures in feet, inches, and eighths of an inch.
- The engineer's rod measures in feet, tenths of a foot, and hundredths of a foot.



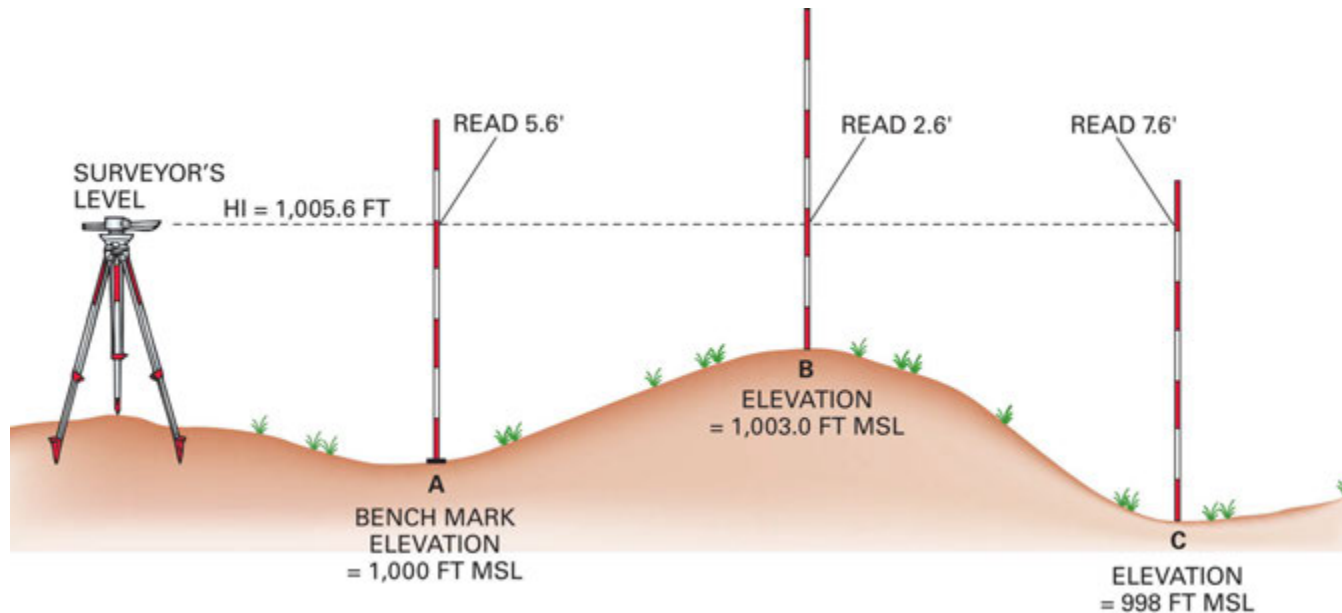
2.1.2 Vertical Distance Measurements (3 of 4)

A leveling rod is placed on a benchmark that has a known elevation, and then it is read by the horizontal **line of sight** from the level instrument after it is first set up and leveled. This reading is the **backsight (BS)** reading.



2.1.2 Vertical Distance Measurements (4 of 4)

The figure below shows how a surveying level and a leveling rod permit calculating grade elevations.



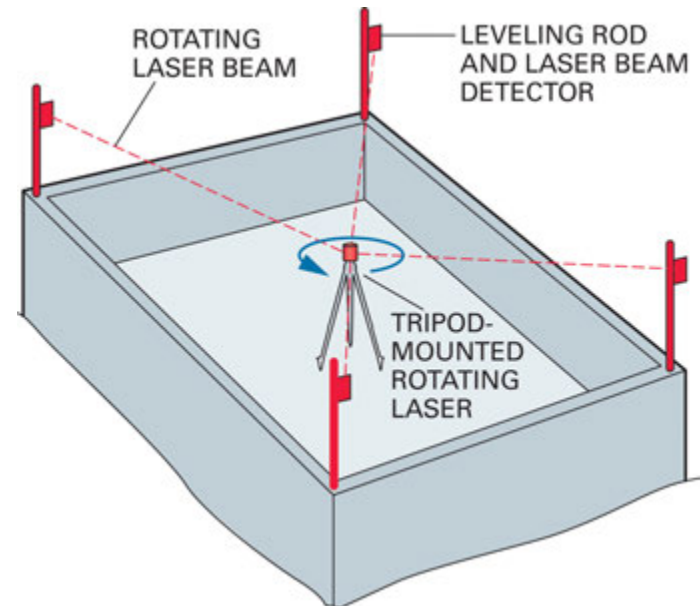
2.2.0 Electronic Equipment and Systems (1 of 2)

- Laser levels provide users with increased precision and a digital readout of measurements.
- Newer laser levels permit one-person operation.



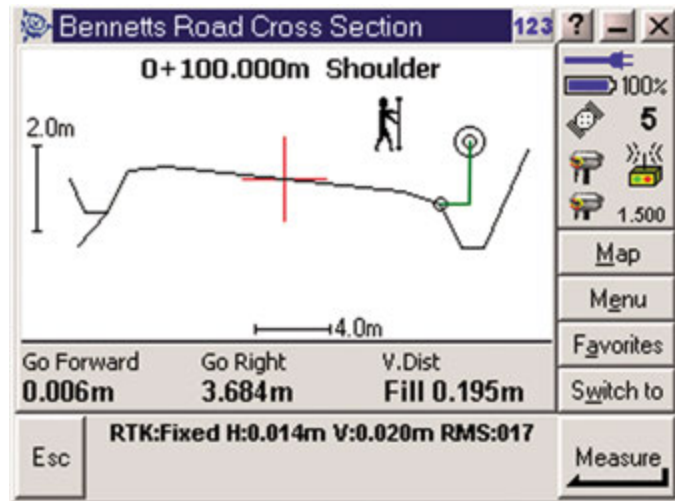
2.2.0 Electronic Equipment and Systems (2 of 2)

- The single-beam laser projects a line of light (visible or invisible) towards a receiver.
- Rotating beam lasers sweep a beam of light over a 360-degree area so that multiple receivers can be placed around the site.



2.2.1 Global Positioning Systems

A GPS receiver may be placed at any convenient location on the work site and a benchmark may be quickly and accurately set.



2.2.2 Stakeless Systems

- Computer technology now allows surveyors to lay out construction sites electronically on a computer that converts the layout to transmittable signals.
- Signal receivers mounted on the equipment such as dozers and motor graders can receive these signals.
- Computers calculate the location of the equipment, with or without GPS, and match it to the location on the construction plans.



2.3.0 Establishing and Checking Finish Grades

- After completion of rough-grading work, the survey crew sets finish-grade stakes.
- These stakes are set at exact levels and provide a visual check of the final level of the subgrade and grade.
- On small jobs, the survey crew may set finish grade stakes at the beginning of the job. If this is the case, equipment operators must be aware of how they are set and be sure to complete the grading work correctly.



2.3.1 Finish Grade Stakes

- Finish stakes indicate the exact finish of a subgrade or grade.
- Subgrade finish stakes are color-coded red and are commonly called red tops.
- Grade finish stakes are color-coded blue and have the name blue tops.
- The top of the finish grade stake is set level with the grade or subgrade, or the top of the stake is set and marked at a predetermined level above the grade or subgrade.



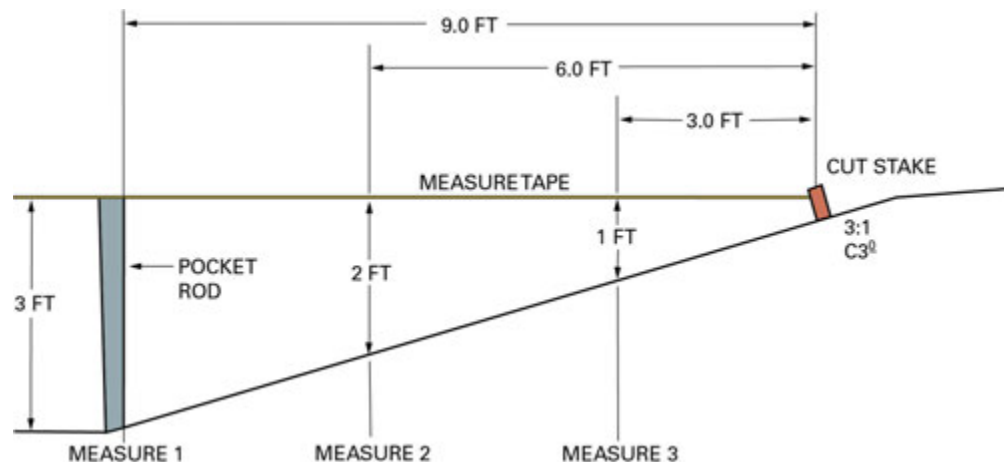
2.3.2 Checking Grades (1 of 4)

- Three people are normally needed to manually check subgrades and grades.
- One person, using various types of equipment, can complete the work.
- A grade can be checked roughly using a pocket leveling rod, level, and tape measure.



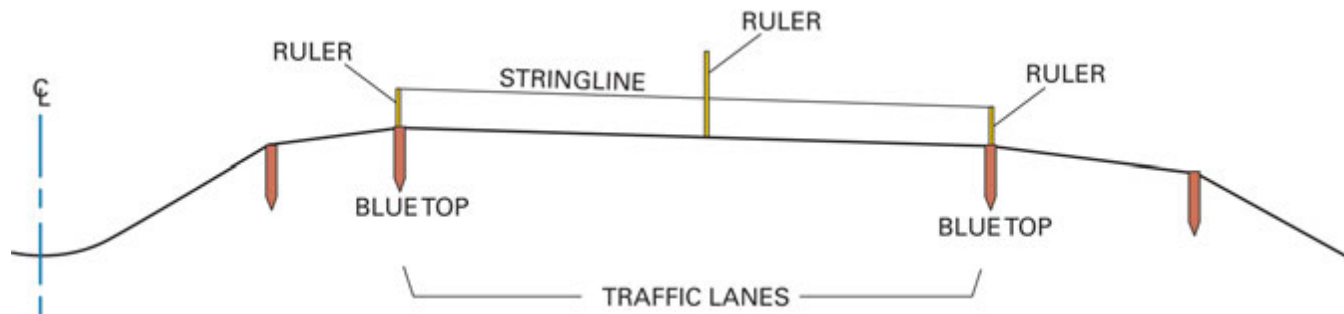
2.3.2 Checking Grades (2 of 4)

1. Attach the tape measure to the cut stake and reel it out to the point indicated on the cut stake. In this case, it should be 9.00 feet away.
2. Hold the pocket leveling rod vertically on the ground.
3. Hold the tape horizontally against the rod, using the level to ensure that the string is level. In this case, the rod should read 3.00 feet.



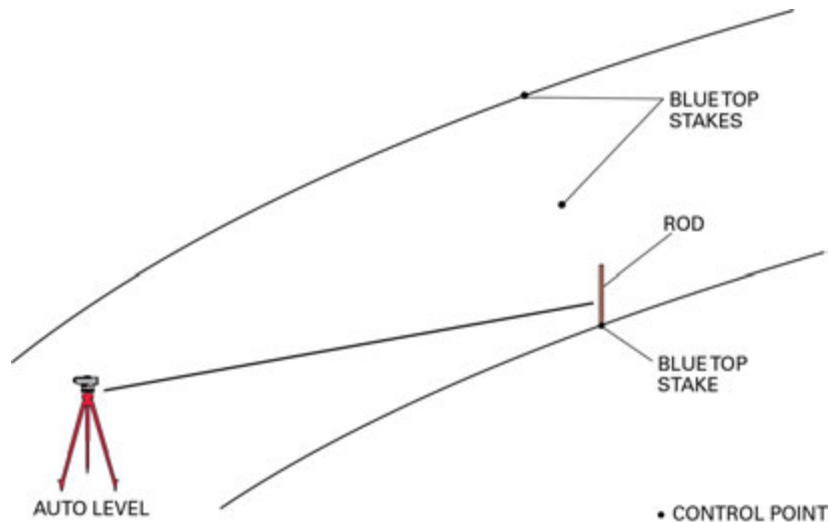
2.3.2 Checking Grades (3 of 4)

1. Check the level of the surface. Stretch the string perpendicular to the center line across the roadway. Two people set a ruler on top of the finish stakes and hold them at the same level.
2. Check the grade of the surface. The third person randomly measures points across the roadway to ensure that the surface is the same height between the subgrade and string.



2.3.2 Checking Grades (4 of 4)

- An automatic level or laser level and rod, or a laser level and detector, may be used to check the grade of the subgrade.
- Both methods require the measurement of a known elevation (control point).
- Use GPS technology whenever possible, since it provides the most accurate readings.



Wrap Up – Trade Terms (1 of 2)

Backsight (BS): A reading taken on a leveling rod held at a known elevation to determine the elevation of the leveling instrument (height of instrument) after it is first set up and leveled.

Foresight (FS): A reading taken on a leveling rod to determine an unknown elevation at the location of the leveling rod; used in conjunction with a known height of instrument.

Global positioning system (GPS): System developed and maintained by the US military to provide precise information for the movement of military vehicles, equipment, and personnel. This system is now available for use by anyone and can provide precise location and elevation data.



Wrap Up – Trade Terms (2 of 2)

Height of instrument (HI): In surveying, the geographical elevation above sea level or other reference of a level instrument's line of sight. Equal to the benchmark elevation plus backsight height obtained with a leveled instrument.

Line of sight: The imaginary line passing from the eye along the optical axis of the instrument through the cross hairs of the reticle to the target of interest.

Mean sea level (MSL): The average vertical position of the actual or assumed surface of the sea at any point on the globe, based on a geometric model of the earth's shape. Generally used as the zero-reference for elevation.



2.0.0 Section Review Question 1

1. The line of sight in a surveyor's level during an observation must be _____.
- a. vertical
 - b. parallel to grade
 - c. horizontal
 - d. at sea level



2.0.0 Section Review Question 1 Answer

1. The line of sight in a surveyor's level during an observation must be _____.
a. vertical
b. parallel to grade
c. horizontal
d. at sea level



2.0.0 Section Review Question 2

2. A rotating beam laser's main advantage over manual levels is that _____.
- a. multiple leveling-rod readings can be taken at the same time
 - b. they are safer to use
 - c. the equipment costs less than manual equipment
 - d. anyone can use the equipment



2.0.0 Section Review Question 2 Answer

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2.0.0 Section Review Question 3

3. While checking a finish grade, you note that a cut stake instruction reads: F/S, 54^6 in a circle, 4:1, followed by F 5^{00} @ 20^{00} . What will the cut depth be 10' from the stake toward the center line?
- a. 2.00'
 - b. 2.50'
 - c. 5.00'
 - d. 10.00'



2.0.0 Section Review Question 3 Answer

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Next...



Review and Module Examination

Complete the Review Questions at the end of the module and prepare for the Module Exam.